



Remote Intern Challenge

FX Risk Alert & Currency Dashboard

Welcome to your remote team challenge.

Your mission is to build a small finance/FX system that monitors exchange rates, analyzes currency movements, and helps a user understand when a currency becomes risky or volatile.

The objective is not to build a perfect product. The objective is to see how you work together as a team, how you organize tasks, how you handle data, and how you transform financial data into a useful software idea.

Project Scenario

Imagine you are building a simple dashboard for a company that follows foreign exchange markets.

The company wants to monitor currencies such as EUR/USD, USD/TND, EUR/TND, GBP/USD, or any other available currency pair.

The system should help answer questions such as:

- Is the currency moving up or down?
- Which currency is the most volatile?
- Did the exchange rate move strongly today?
- What was the highest and lowest level over a selected period?
- Should the system show a warning when movement is unusual?
- How can this information help a business exposed to foreign currency?

What You Need to Build

Your team should create a small system with:

1. Data research
2. Data cleaning
3. Database
4. Backend
5. Frontend dashboard
6. FX analysis
7. Final presentation

1. Dataset Research

Find a public dataset related to foreign exchange rates.

You can use Kaggle, Hugging Face, GitHub, or another open-source platform.

In your submission, explain:

- Which dataset you selected
- What columns it contains
- What currencies are available
- Why this dataset is useful
- What business questions it can answer

2. Data Cleaning

Clean the dataset before using it.

You should handle:

- Missing values
- Duplicate rows
- Wrong date formats
- Wrong number formats
- Column renaming
- Invalid exchange rates

- Basic consistency checks

At the end, you should have a clean dataset ready for analysis.

3. Database

Create a simple database to store the clean data.

You can use:

- SQLite
- PostgreSQL
- MySQL
- Supabase
- Firebase
- Any other simple database tool

Example tables:

- currencies
- exchange_rates
- daily_performance
- alerts
- optional_news_sentiment

4. Backend

Build a simple backend API.

The backend should allow the frontend to request:

- List of available currencies
- Historical exchange rates
- Daily percentage change
- Weekly or monthly performance
- Highest and lowest exchange rate
- Volatility measure
- FX risk alert

- Optional news sentiment result

You can use Flask, FastAPI, Django, Node.js, or any backend framework.

5. Frontend Dashboard

Build a simple dashboard where the user can select a currency and see useful information.

The dashboard should include:

- Currency selector
- Historical exchange rate chart
- Daily change
- Weekly or monthly change
- Highest and lowest rate
- Volatility indicator
- Risk alert: low, medium, or high
- Simple explanation of the currency movement

Example dashboard cards:

- Current rate
- 1-day change
- 7-day change
- 30-day change
- Highest level
- Lowest level
- Volatility level
- Risk status

6. FX Analysis

Your system should produce useful analysis, not only charts.

You should answer:

- Which currency moved the most?

- Which currency was the most stable?
- Which currency was the most volatile?
- During which period did the strongest movement happen?
- What could this mean for a company that imports or exports?
- When should the system show an alert?

7. Risk Alert Logic

Create a simple alert system.

Example:

- Low risk: daily movement below 0.5%
- Medium risk: daily movement between 0.5% and 1%
- High risk: daily movement above 1%

You can create your own logic, but you must explain it clearly.

The alert does not need to be perfect. It only needs to be logical and useful.

8. Optional AI Feature

If your team has time, add one AI or automation feature.

Examples:

- Automatic FX market comment
- Currency movement summary
- Simple chatbot that answers questions about the dashboard
- News sentiment section
- Automatic alert explanation

Example:

Input:

USD/EUR moved by 1.2% today and volatility increased.

Output:

“The currency pair showed a strong daily movement, suggesting higher short-term FX risk. A company exposed to this currency should monitor pricing and payment timing carefully.”

Final Deliverables

Your team should submit:

1. A short presentation
2. A working prototype or demo
3. The cleaned dataset
4. The database structure
5. Backend code
6. Frontend/dashboard
7. A short explanation of your FX analysis
8. A list of future improvements

Suggested Team Roles

You can divide the work like this:

- Project coordinator: organizes the team and tracks progress
- Data researcher: finds and explains the dataset
- Data analyst: cleans data and prepares analysis
- Backend developer: builds the API and database connection
- Frontend developer: builds the dashboard
- Presenter: prepares the final demo and explanation

Each member should explain clearly what they contributed.

Evaluation Criteria

Your work will be evaluated based on:

- Team organization
- Quality of data cleaning
- Database structure
- Backend logic
- Frontend clarity
- Quality of charts

- Quality of FX analysis
- Practical value of the risk alerts
- Creativity
- Final presentation
- Ability to explain technical and business choices

Important Note

The goal is not to build a perfect financial platform.

The goal is to see how you work as a team, how you divide tasks, how you solve problems, and how you transform raw FX data into a useful software idea.